

Department of Mechanical Engineering
B.Tech Project Presentation on

VOLLEY: Controlled CubeSat Deployment from a Hosted Orbital Platform

Completed Gen5 system-level computational evaluation



Presented By

Adityavardhan Mishra · Pratham Chawla

PRN 23070125054 · PRN 23070125029

Guide: Vikas Gulia

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SYMBIOSIS INSTITUTE OF TECHNOLOGY PUNE

Presentation route

OUTLINE

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The thesis answers an orbital-delivery trade question

INTRODUCTION

Shared destination

Secondary CubeSats inherit the orbit chosen for the primary customer.

Conventional release

Spring deployers provide a small relative impulse. Subsequent orbit or drag differences can produce phasing.

VOLLEY question

Can commanded release provide useful orbital choice at an acceptable installed cost?

Study outcome: useful modelled orbital change, but an unfavourable 3U mass trade

Existing methods solve different parts of the problem

LITERATURE REVIEW

Approach	Strength	Limit for this thesis
Spring deployer	Flight heritage; compact; simple	Limited tailored release speed
Spring release / drag	Relative state or drag can create phasing	Timing alone, with no relative state change, cannot
Orbital transfer vehicle	Larger orbit changes	Greater bus, mass and integration burden
Electromagnetic research	Commandable acceleration studied	Armature, high-g or maturity constraints vary

A narrower, testable contribution

RESEARCH GAP

Commanded release for an unmodified CubeSat at moderate acceleration, with the full system counted.

Contribution examined

Gen5 reusable-sled linear motor, 3U payload, magazine, pulse power, arrest and orbital utility.

Boundary

Electromagnetic CubeSat deployment exists in prior work. Payload qualification and physical performance remain unproved.

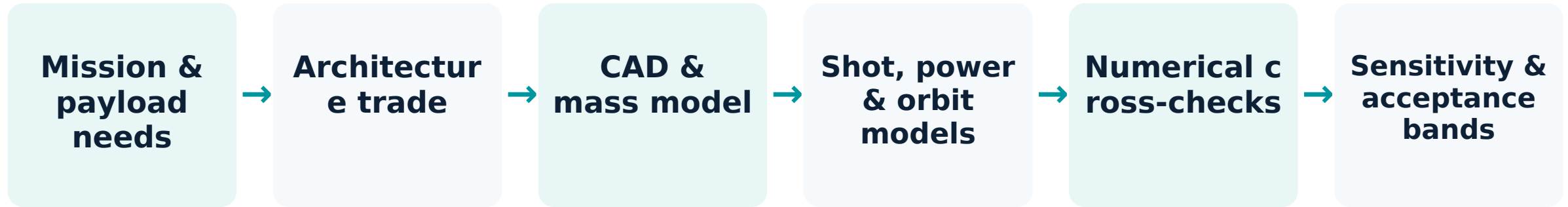
The analytical research objectives have outcomes

OBJECTIVES

Research objective	Final thesis outcome
Design a commandable 3U system	Gen5 CAD and complete subsystem architecture documented
Calculate shot and control performance	Rated shot, power and dispersion models completed
Evaluate orbital utility	Predicted energy and lifetime benefit for stated cases
Count full installed burden	Mass rollup completed; 3U parity criterion failed
Test credibility of key claims	Numerical cross-checks and limits recorded

A completed analysis chain from mission to decision

RESEARCH METHODOLOGY



Models were challenged and revised; hardware performance remains outside this thesis's evidence.

Numerical checks strengthened the final baseline

RESEARCH METHODOLOGY

Question	Method / cross-check	Evidence limit
Field and thrust	Analytic winding; magpylib; FEM	No measured thrust constant
Shot and power	Dynamics; Monte Carlo; ngspice	Assumed components and interfaces
Structure and flow	CAD; CalculiX; OpenFOAM	No payload qualification
Orbital change	Orbit model; Cowell / GMAT cases	Lifetime depends on atmosphere

Project evolution: documented decisions and analysis

TIMELINE

2023

Hosted concept Free-flyer concept reframed around a host platform

Mid-2025*

LSM direction Coilgun study gives way to a linear synchronous motor

2025-26*

CAD generations Geometry and mass models mature through Gen5

Jul-Sep 2026

Gen5 evidence Cross-checks, defect corrections and thesis record

Oct 2026

Final review Gen5 results, failed criterion and thesis record presented

* Approximate historical milestone; not a documented semester deadline.

2026 findings were incorporated in the final baseline

TIMELINE

Jul

CAD-derived sled mass replaced an optimistic parametric estimate

Aug

Depth-resolved thrust calculation lowered the rated speed

Aug

Enclosure buildup exposed the full 3U mass penalty

2026

Independent orbit checks rejected a solar-invariance claim

Final

Results and failures retained in the submitted thesis record

The thesis deliverables are complete and traceable

WORK PROGRESS

Design delivered

Gen5 CAD, magazine and release architecture, subsystem models and system mass rollup.

Analysis delivered

Shot, control, circuit, structure, thermal, orbit and alternative-case results.

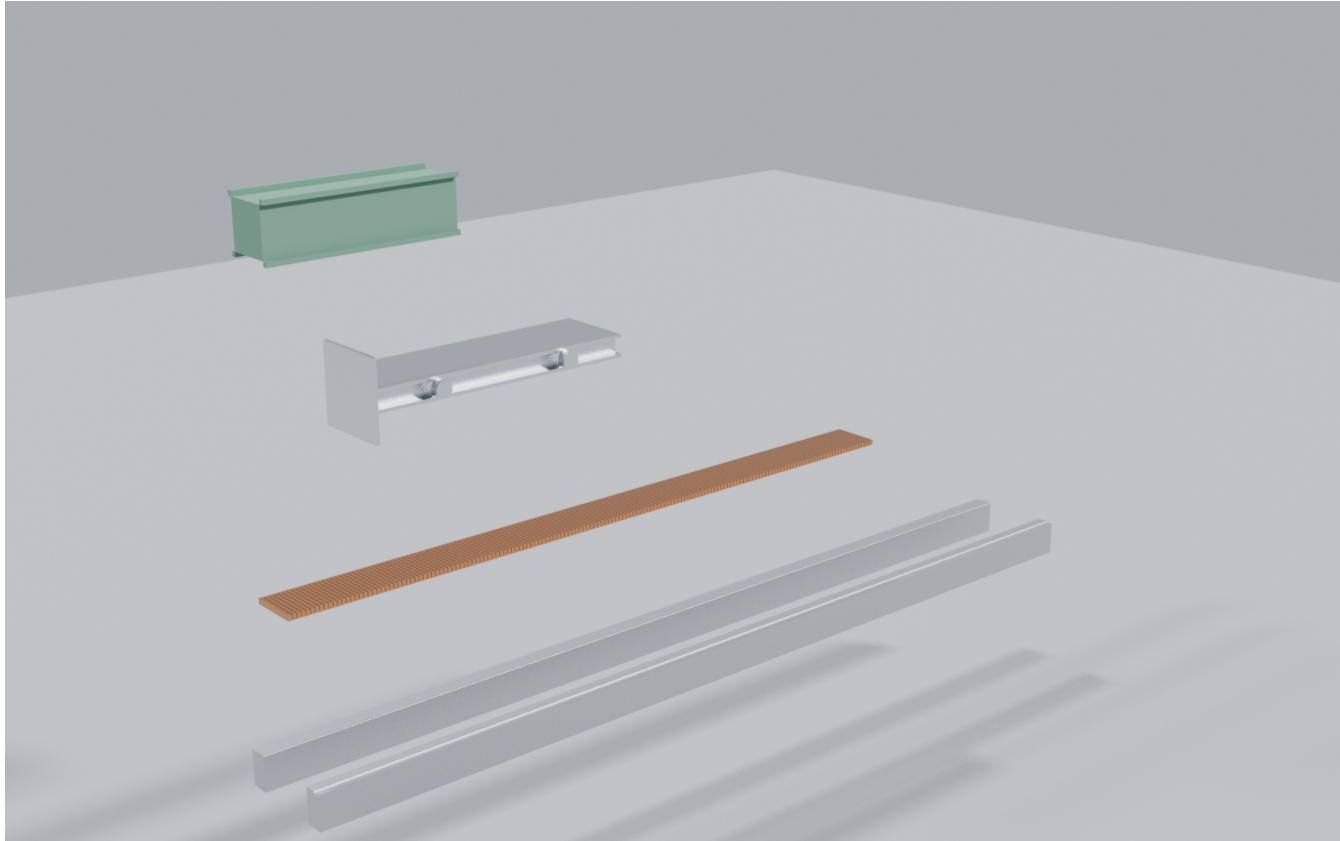
Evidence delivered

Manuscript, figures, scripts, numerical checks, provenance and defect register.

Scope: completed computational thesis; physical qualification belongs to a later development phase.

Gen5: complete computational system configuration

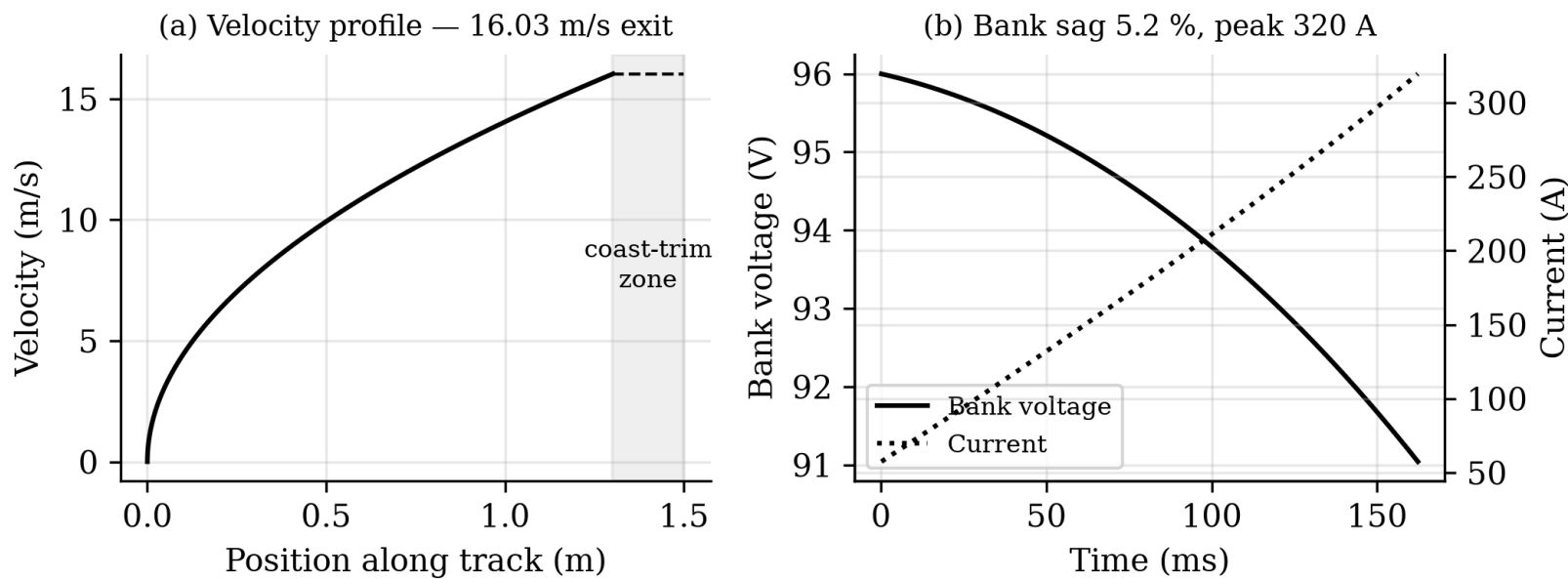
RESULTS



- 1.5 m linear-motor track and double-sided Halbach arrays
- Reusable 9.45 kg magnet sled
- Two cassettes; twelve 3U satellites
- Pulse bank, restraint and eddy-current arrest

Rated 3U shot: performance predicted by the model

RESULTS



16.0 m/s
exit speed

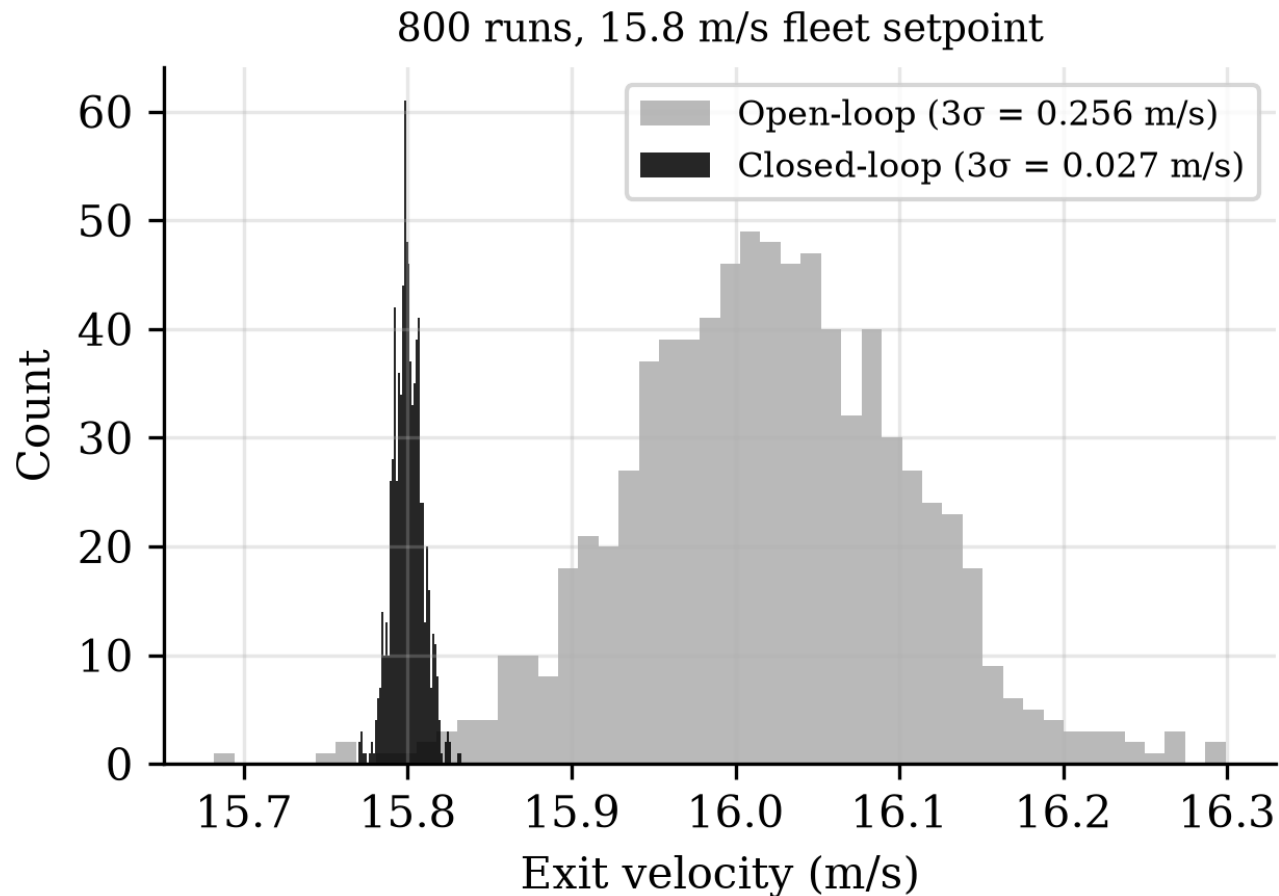
10.07 g
acceleration

2.782 kJ
gross draw

18.8%
net efficiency

Modelled command precision is narrow; it is unmeasured

RESULTS



0.0274 m/s

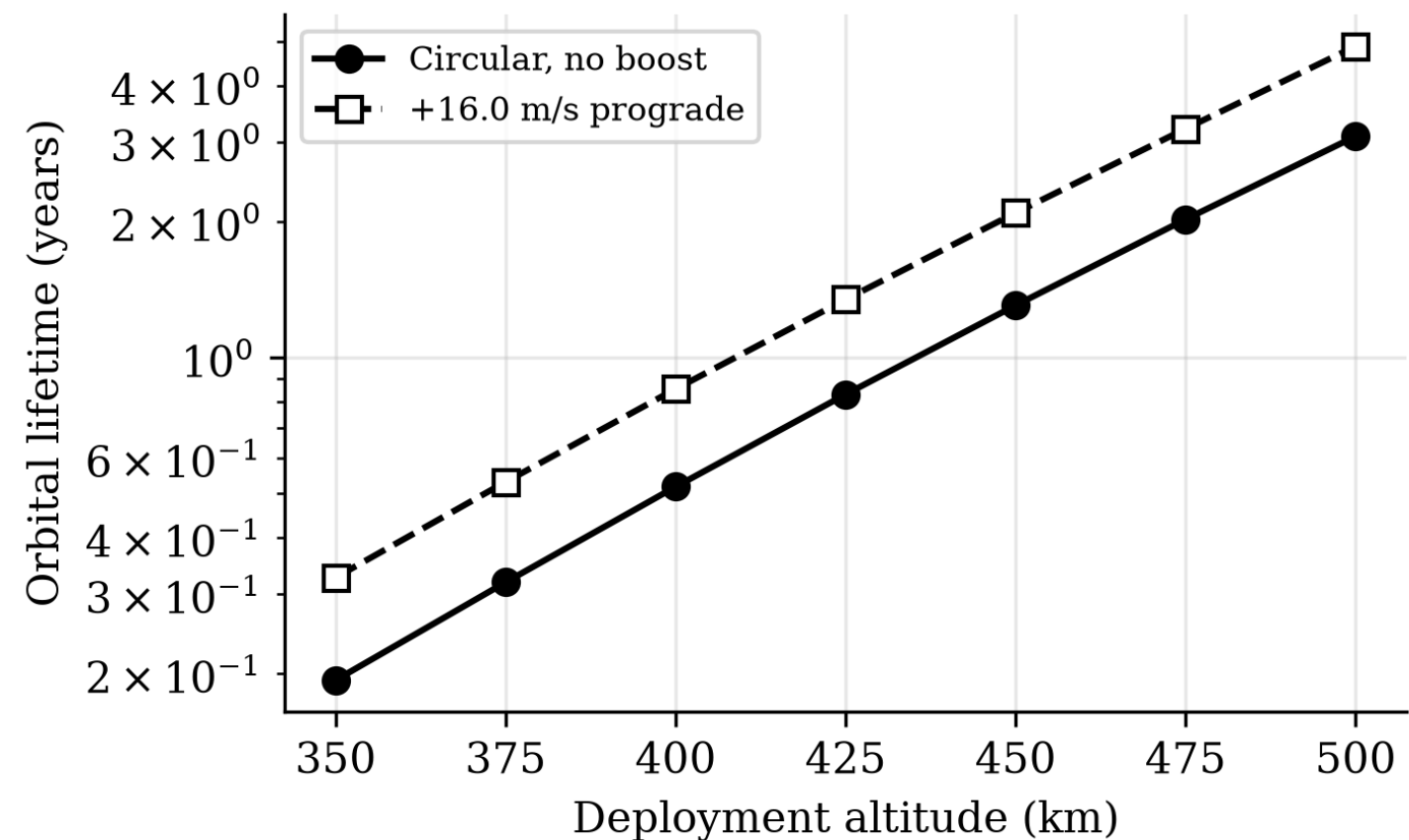
3σ exit-velocity dispersion at a 15.8 m/s fleet setpoint.

Evidence limit

Monte Carlo under modelled sensing and plant tolerances; no measured dispersion.

Commanded velocity can change orbital energy

RESULTS



+28.8 km

Predicted semi-major-axis rise for one maximum-velocity release.

1.60×

Predicted lifetime at stated mean solar activity; atmosphere-dependent.

Persistent phasing requires a relative state change; this result concerns orbital energy.

Gen5 fails its 3U mass comparison

RESULTS

Modeled dry mass, 12 slots **126.6 kg**

Gen5 per 3U  **10.55 kg**

Spring comparator  **~6 kg**

1.76× heavier per 3U payload; the preset 15% parity band fails.

The study ends with a clear engineering decision

RESULTS

Supported

Modelled commandable release and a conditional orbital-energy benefit.

Rejected for 3U

The complete Gen5 configuration misses the preset spring-mass parity target.

Unverified

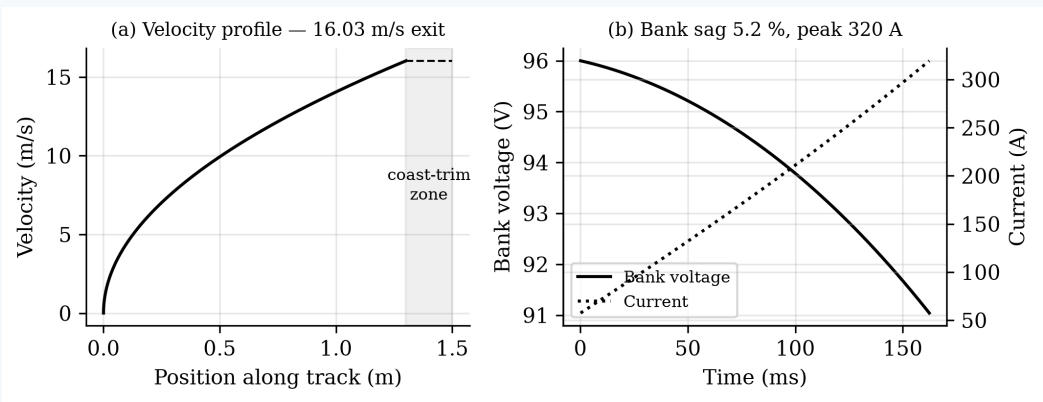
Payload loads, magnetic cleanliness, cycle life and host interfaces require tests.

Decision: retain the numerically checked model; do not select Gen5 as a 3U flight configuration.

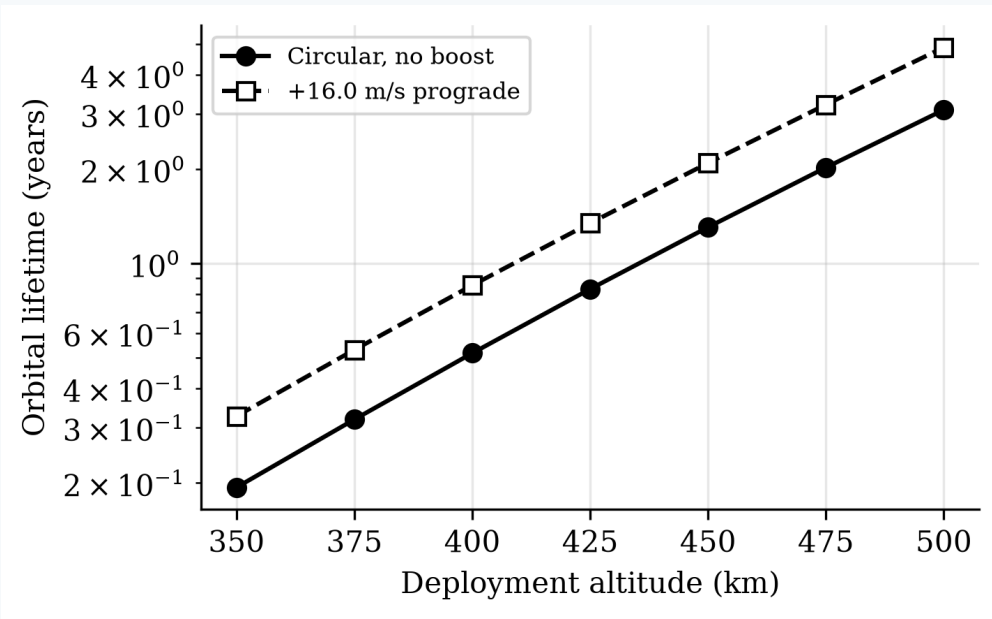
Computational demonstration: from shot to orbit

RESULTS

1. Simulate a 3U shot → 16.0 m/s



2. Model the orbital response



**Compare predicted orbital change and full installed mass with the spring baseline.
Simulation, not physical demonstration.**

The research question has a completed answer

CONCLUSION

- The Gen5 system-level model predicts 16.0 m/s 3U release and a conditional orbital benefit.
- The complete modelled system is 76% heavier per 3U customer than the spring comparator.
- Independent numerical checks corrected the baseline and defined the limits of its claims.
- Conclusion: the analytical objective is complete; Gen5 does not pass the 3U selection criterion.

Gen6 reopens the mechanism and installed-system trade

FUTURE WORK

One shared path • sequential feed • commanded release for each CubeSat

- Investigate approximately 1–2 m/s through 1 km/s across mission cases.
- Compare electromagnetic, stored-energy, gas/fluid, hybrid, BOLLEY and conventional options.
- Count feeder, retention, arrest, control, energy, thermal, host loads and shared faults.

1 km/s is a research target, not demonstrated speed or payload/provider compatibility.

Next milestone: one reviewable first-test configuration

FUTURE WORK

Freeze

Named mission, payload and host surrogate

Decide

Fair mechanism and installed-burden trade

Measure

Field, contact, release, arrest, loads and cycle behavior

Review

Drawings, BOM, procedures, uncertainty and review decision

Core literature and standards (1/2)

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Methods and project records (2/2)

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